Java while loop

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In programming while loop plays vital role in iterating over a set of statements repeatedly. until a specific condition is met. One such loop in Java is the while loop known for its simplicity and versatility.

The Java while loop is used to iterate a part of the program repeatedly until the specified Boolean condition is true. As soon as the Boolean condition becomes false, the loop automatically stops.
The while loop also considered as repeating if a statement. If the number of iteration is not fixed, it is recommended to use the while loop.

Syntax:

```
while(condition){
   //code to be executed
}
```

Here condition is a Boolean expression that determines whether the loop should continue iterating or not. The statements within the curly braces are executed repeatedly, as long as condition evaluates to true.

Example:

```
package Looping;
import java.util.*;


/**
 * Write a description of class DemoOfWhileLoop here.
 * Wrting code to input a number, and we will find the sum of
 * all the natural number upto that number.
 * For example:
 * Input:
 * Enter any number: 5
 * Output:
 * 1 + 2 + 3 + 4 + 5 = 15
 * @author (your name)
 * @version (a version number or a date)
 */
public class DemoOfWhileLoop
{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("\f");
        int num, start = 1, sum = 0;
        System.out.print("Enter any natural number: ");
        num = sc.nextInt();
        while(start < num){
            System.out.print(start + " + ");
            sum = sum + start;
            start++;
        }
        System.out.print(start+" = " + (sum + start));
    }
}
```

num = 5
start = 1 2 3 4 5
sum = 0 1 3 6 10

Output:
1 + 2 + 3 + 4 + 5 = 15

10 + 5

Looping in real life

while (switch is on) {
→ Fan has mechanism
if (switch is off) {
→ break the loop.
}
}

Output:

```
BlueJ: Terminal Window - ClassIX
Options
Enter any natural number: 6
1 + 2 + 3 + 4 + 5 + 6 = 21
```

```
Enter any natural number: 6
1 + 2 + 3 + 4 + 5 + 6 = 21
```

Factorial:
5! = 5 * 4 * 3 * 2 * 1 = 120

N! = N * (N-1) * (N-2) * (N-3)* … * 1 (must be '1')

$5! = 5 \times (5-1) \times (5-2) \times (5-3) \times (5-4)$
$= 5 \times 4 \times 3 \times 2 \times 1$

WAP to input the value of N and find the factorial of N.
For example:
Input: 5
Output:
5! = 5 * 4 * 3 * 2 * 1 = 120

```
package Looping;
import java.util.*;


/**
 * Write a description of class Factorial here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Factorial
{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("\f");
        int num;
        double f = 1;
        System.out.print("Enter any natural number: ");
        num = sc.nextInt();
        System.out.print(num + "! = ");
        while(num > 1){
            System.out.print(num + " * ");
            f = f * num;
            num--;
        }
        System.out.print(num + " = " + (long)(f));
    }
}
```

Why?

num = 5 4 3 2 1
f = 1 5 20 60 120

Output:
5! = 5 * 4 * 3 * 2 * 1 = 120

false

true

Mechanism is on,

Why?

while (car is ON) { — condition

(car's mechanism is on.
if (destination is reached) {
break, or stop the car.
}
}

WAP to print all the numbers which is divisible by 2 or 3 or 5, in a range of numbers:

Enter lower bound: 10
Enter upper bound: 100

Numbers divisible by 2 or 3 or 5 are:

10, 12, 14, 15, 16, 18, 20, 21, 22, 24, 25, 26, 27, 28, 30, 32, 33,

numbers divisible by 2 or 3 or 5 are:

10, 12, 14, 15, 16, 18, 20, 21, 22, 24, 25, 26, 27, 28, 30, 32, 33, 34, 35, 36, 38, 39, 40, 42, 44, 45, 46, 48, 50, 51, 52, 54, 55, 56, 57, 58, 60, 62, 63, 64, 65, 66, 68, 69, 70, 72, 74, 75, 76, 78, 80, 81, 82, 84, 85, 86, 87, 88, 90, 92, 93, 94, 95, 96, 98, 99, 100

```
*/
public class cw_8
{
    public static void main(String args[])
    {
        Scanner sc=new Scanner (System.in);
        System.out.print("\f");
        int start=1;
        int end=10;
        int sum=1;
        while(start<end)
        {
            System.out.print(sum+", ");
            sum+=start;
            start++;
        }
        sum+=start;
        System.out.print(sum);
    }
}
```

Dry Run or Debugging of code

output

1, 2, 4, 7, 11, 16, 22, 29, 37, 46